CS23532 - COMPUTER NETWORKS

Lab Experiments

9. Design a simple topology and configure with one router, two switches and PCs using Cisco Packet Tracer.

1. Build the Topology

1. **Add End Devices**: Navigate to **End Devices** in the bottom-left palette. Drag four **PCs** onto the logical workspace. Place PC0 and PC1 on the left side (LAN 1). Place PC2 and PC3 on the right side (LAN 2).
2. **Add Switches**: Go to **Network Devices** > **Switches**. Choose the **Cisco 2960 switch**. Drag two switches onto the workspace (Switch0 for LAN 1, Switch1 for LAN 2).
3. **Add a Router**: Go to **Network Devices** > **Routers**. Drag a **Cisco 4331 or 2911 Router** (Router0) and place it in the center, above both switches.
4. **Cable the Devices**: Select the **Connections** icon (the lightning bolt) and choose the **Copper Straight-Through cable**.
    - Connect **PC0** (FastEthernet0) to **Switch0** (FastEthernet0/1).
    - Connect **PC1** (FastEthernet0) to **Switch0** (FastEthernet0/2).
    - Connect **PC2** (FastEthernet0) to **Switch1** (FastEthernet0/1).
    - Connect **PC3** (FastEthernet0) to **Switch1** (FastEthernet0/2).
    - Connect **Switch0** (GigabitEthernet0/1) to **Router0** (GigabitEthernet0/0/0).
    - Connect **Switch1** (GigabitEthernet0/1) to **Router0** (GigabitEthernet0/0/1).

2. Configure the PCs

Configure each host with its unique static IPv4 address, Subnet Mask, and Default Gateway:

1. Click on **PC0** to open its configuration panel.
2. Go to the **Desktop** tab and click on **IP Configuration**.
3. Select **Static** and fill in the following details:
    - **IPv4 Address**: 192.168.10.10
    - **Subnet Mask**: 255.255.255.0
    - **Default Gateway**: 192.168.10.1
4. Repeat this exact workflow for the remaining endpoints:
    - **PC1 Configuration:**
        - IP Address: 192.168.10.11
        - Subnet Mask: 255.255.255.0
        - Default Gateway: 192.168.10.1

- **PC2 Configuration:**
  - IP Address: 192.168.20.10
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.20.1
- **PC3 Configuration:**
  - IP Address: 192.168.20.11
  - Subnet Mask: 255.255.255.0
  - Default Gateway: 192.168.20.1

3. Configure the Router Interfaces

By default, Cisco router interfaces are disabled and powered down. You must explicitly assign IP addresses and activate them using the Command Line Interface (CLI):

1. Click on **Router0** and click the **CLI** tab.
2. If prompted with Would you like to enter the initial configuration dialog? [yes/no]:, type **no** and press **Enter**.
3. Execute the following sequential Cisco IOS commands:

```
Router> enable

Router# configure terminal

Router(config)# interface GigabitEthernet0/0/0

Router(config-if)# ip address 192.168.10.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# exit


Router(config)# interface GigabitEthernet0/0/1

Router(config-if)# ip address 192.168.20.1 255.255.255.0

Router(config-if)# no shutdown

Router(config-if)# end

Router# write memory
```

(Note: Once you input the no shutdown command, the physical link lights in your topology diagram will transition from red to green, indicating operational state.)

4. Verify Connectivity

1. Click on **PC0**, go to the **Desktop** tab, and open the **Command Prompt**. 2.

Run a command-line ping to your local gateway to confirm the local link:

ping 192.168.10.1

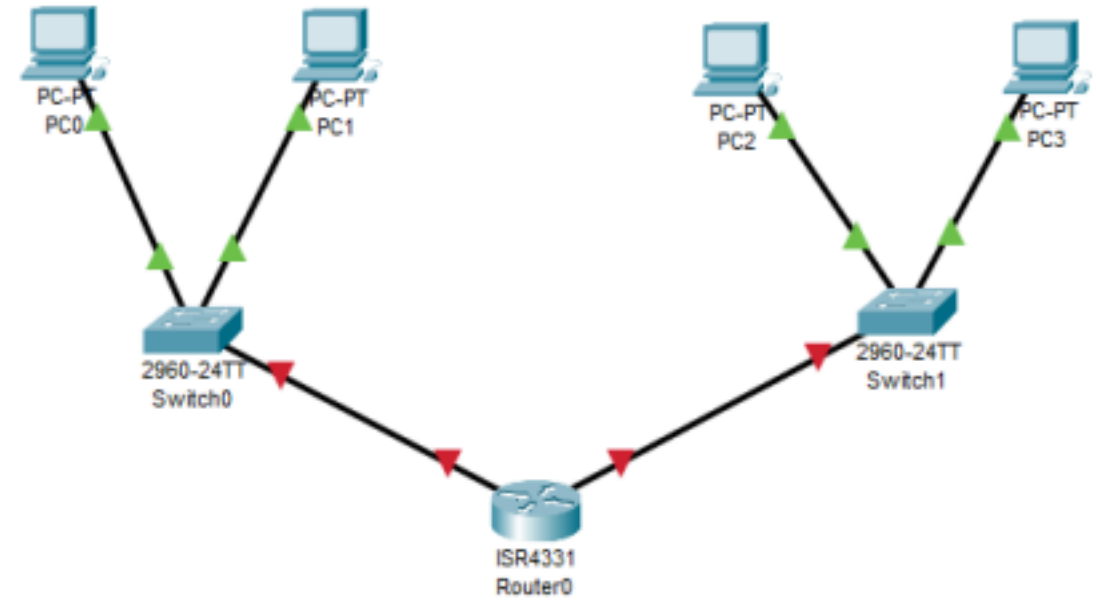
3. Test inter-subnet routing across the network layer by running a cross-network ping to a PC on LAN 2:

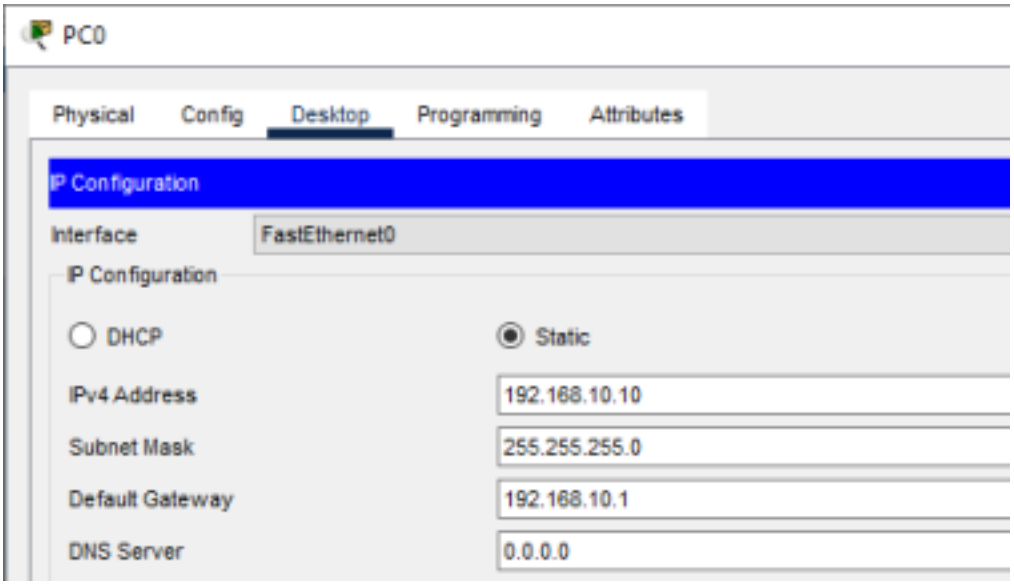
ping 192.168.20.10

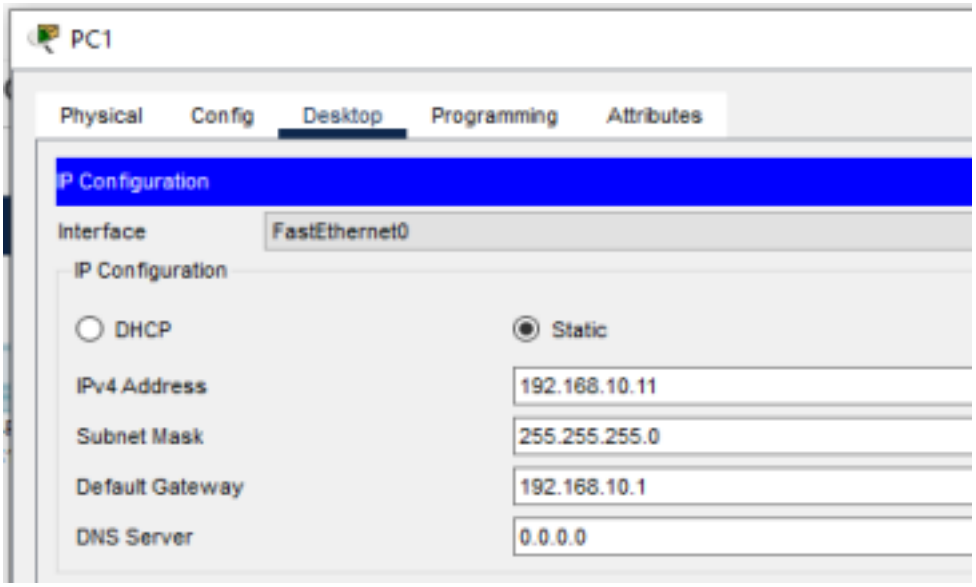
Result

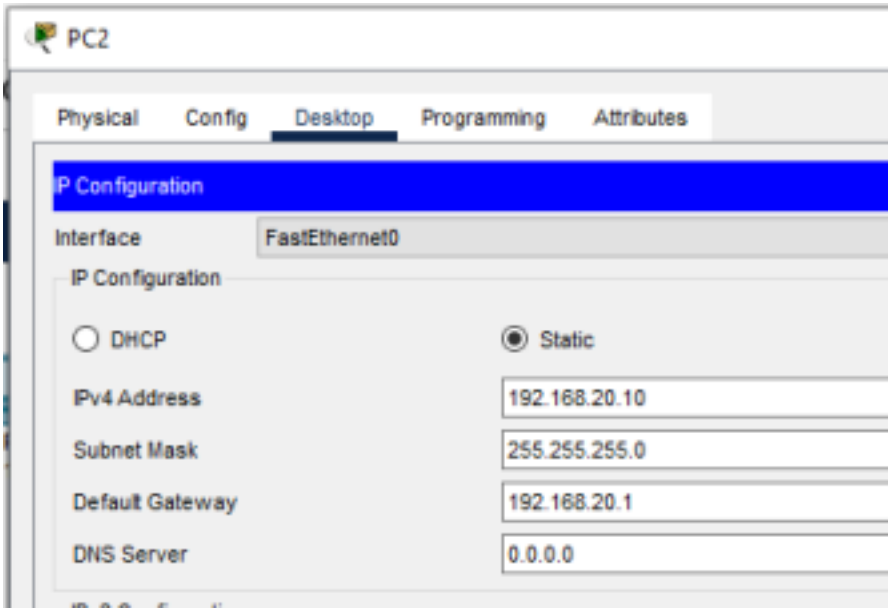
The network topology is operational. The first individual packet packet of an inter-subnet ping may initially time out due to Address Resolution Protocol (ARP) discovery, but subsequent responses will return **Successful** (Reply from 192.168.20.10: bytes=32 time...). The router is managing cross-subnet traffic between LAN 1 and LAN 2.

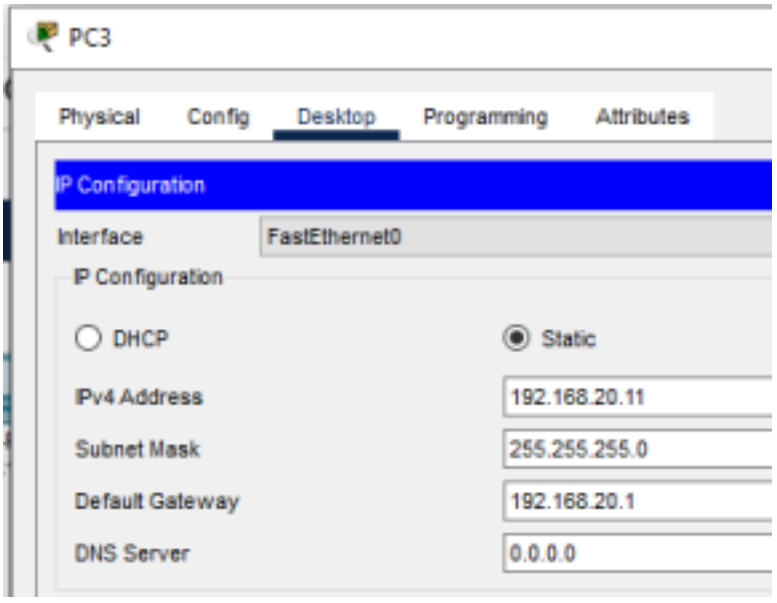
Sample Output

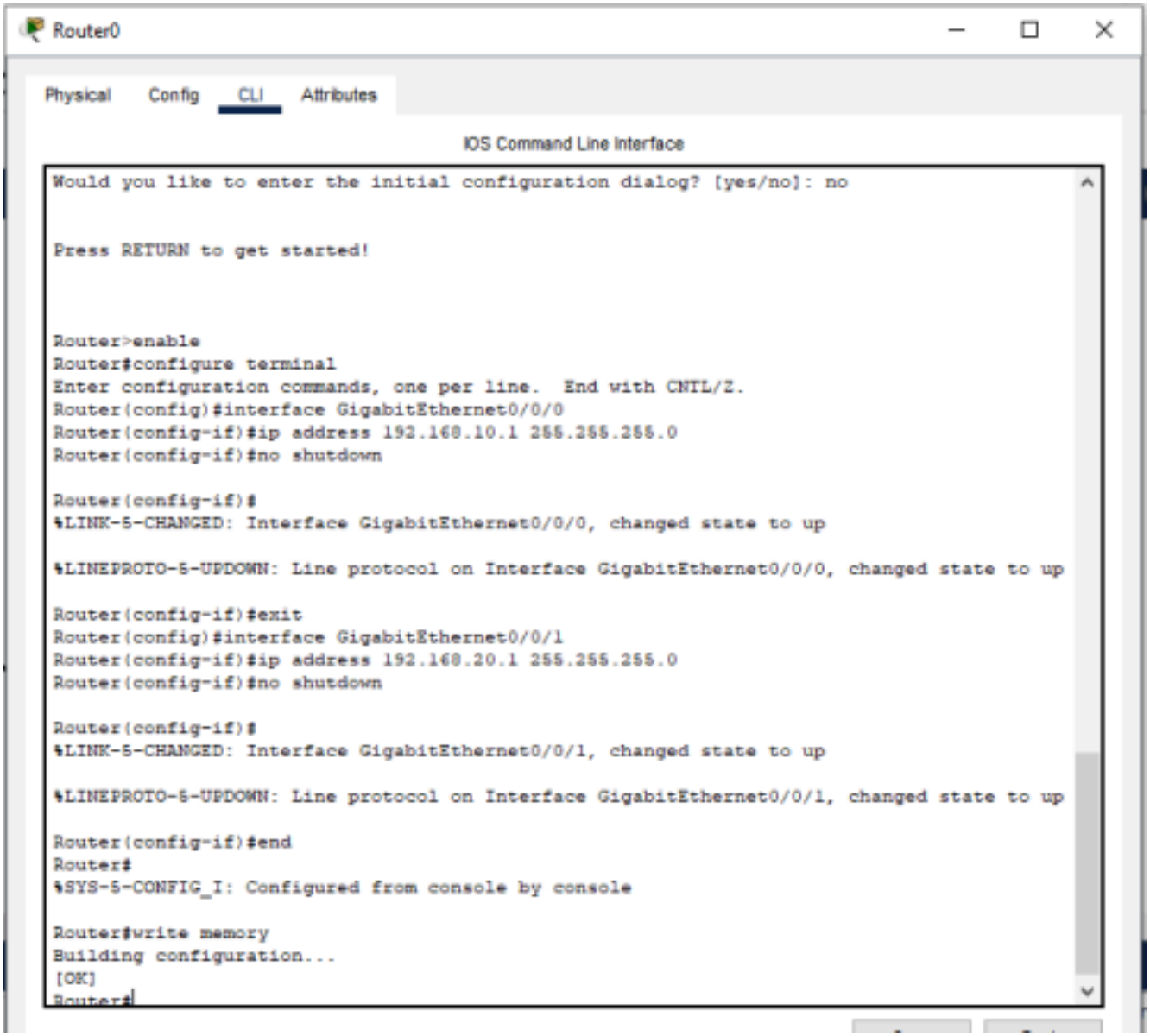


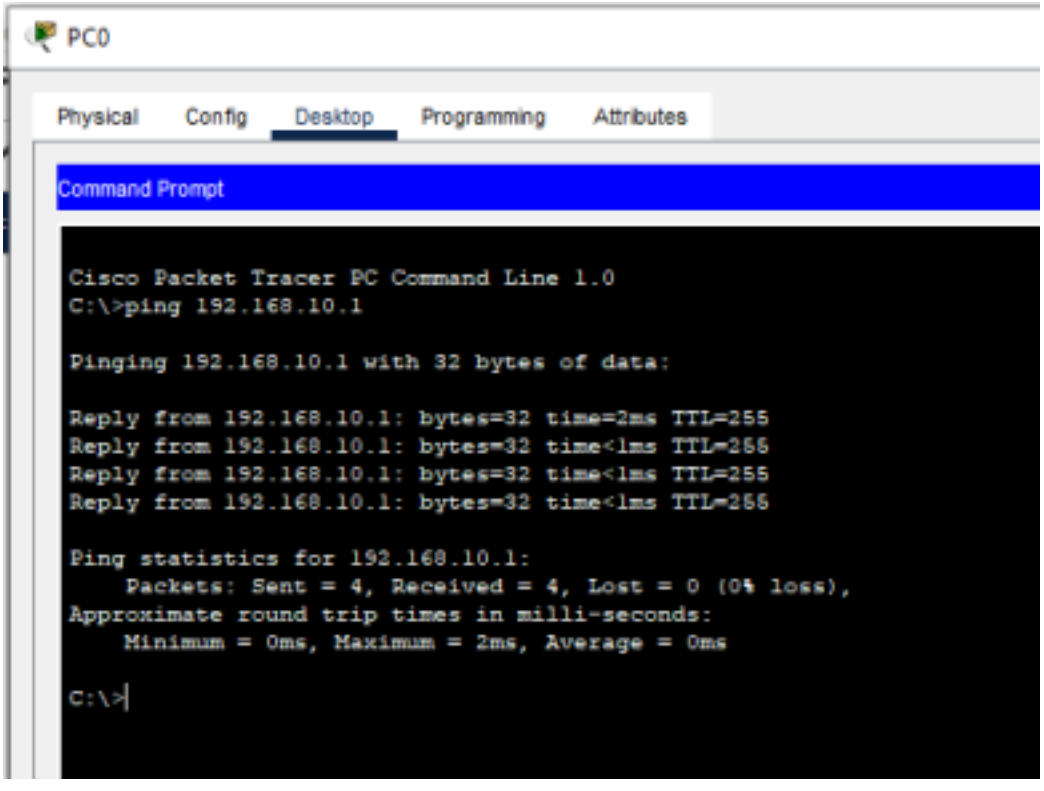


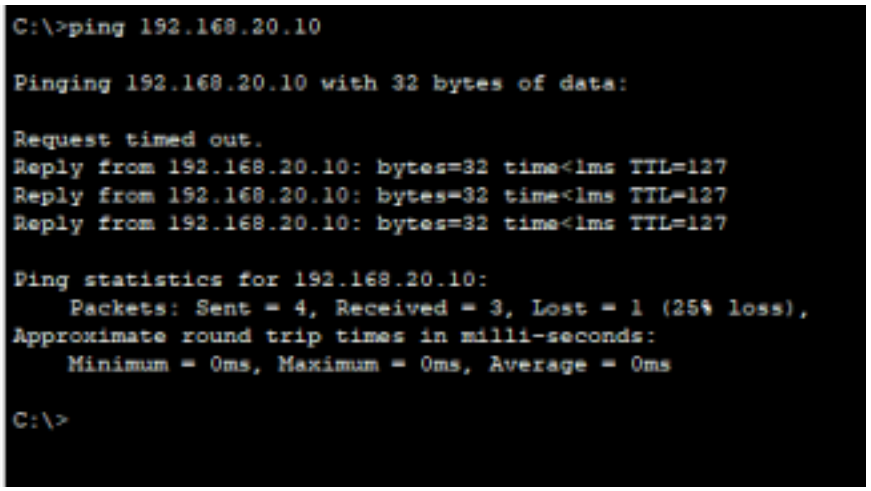

PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
0.0.0.0


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.10
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0


PC3
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.20.11
Subnet Mask
255.255.255.0
Default Gateway
192.168.20.1
DNS Server
0.0.0.0


Router0
Physical
Config
CLI
Attributes
IOS Command Line Interface
Would you like to enter the initial configuration dialog? [yes/no]: no
Press RETURN to get started!
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface GigabitEthernet0/0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/0, changed state to up
Router(config-if)#exit
Router(config)#interface GigabitEthernet0/0/1
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0/1, changed state to up
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
Router#write memory
Building configuration...
[OK]
Router#


PC0
Physical
Config
Desktop
Programming
Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.10.1
Pinging 192.168.10.1 with 32 bytes of data:
Reply from 192.168.10.1: bytes=32 time=2ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255
Ping statistics for 192.168.10.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 2ms, Average = 0ms
C:\>


C:\>ping 192.168.20.10
Pinging 192.168.20.10 with 32 bytes of data:
Request timed out.
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Reply from 192.168.20.10: bytes=32 time<1ms TTL=127
Ping statistics for 192.168.20.10:
Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\>